

Three-dimensional discrete element simulation of electrode structural evolution in lithium-ion batteries during drying and calendaring

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ABSTRACT

The drying and calendaring processes are critical in the manufacture of electrodes for lithium-ion batteries, and have a profound effect on their mechanical and electrochemical properties. In this study, we developed a three-dimensional representative volume element (RVE) model of electrodes, which includes the active material, carbon binder domain, solvent, and particle contacts. Utilizing the discrete element method (DEM), we continuously simulated the structural evolution of the RVE during the drying and calendaring processes. Our simulations revealed a three-stage drying scheme consistent with experimental observations, demonstrating the accuracy of our DEM-based approach. Furthermore, we found that the calendaring process significantly enhances the mechanical integrity and electronic conductivity of the electrodes, with peak stresses occurring in the thickness direction. This research underscores the potential of DEM in elucidating electrode heterogeneity during manufacturing processes and highlights the innovative use of this method in the field of battery science.

1. Introduction

With the advantages of high energy density, high power density and long cycle life, lithium-ion batteries (LIBs) have been widely used in portable electronics, electric vehicles [1,2] and battery energy storage systems [3]. LIBs electrode manufacturing process is complex [4], and it plays a crucial role in advancing LIBs technologies, as it has a significant influence on energy density, manufacturing cost, and throughput [5,6]. Furthermore, drying and calendaring, the most important of the electrode manufacturing processes, are time and energy-consuming and account for over 50 % of the entire electrode processing cost [7,8]. As these processes significantly affect the electrochemical performance and mechanical properties of LIBs [9,10], it is important to understand the evolution of the macroscopic and microscopic structures of electrodes during the drying and calendaring processes.

Several researchers have experimentally investigated the drying process, with emphasis on the effect of drying on the electrode structure and component distribution [11,12]. Sahore et al. [13] investigated the cracking behavior of thick water-based $\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ (NCM811) cathodes during drying and found that the ability of the electrode to withstand drying stress can be improved by modifying the electrode microstructure. Klemens et al. [14,15] comparatively investigated the

drying behavior of NCM cathodes with compact (NCM-C) and porous (NCM-P) structured particles as well as the effect of the layer configuration on the electrode performance. They found that compared with slurries containing NCM-C particles, those containing NCM-P particles exhibit a more uniform binder distribution during a faster drying process. Niesen et al. [16] investigated the effect of drying temperature on the distribution of the binder, which accumulates on the electrode surface at higher temperatures. Jaiser et al. [17,18] experimentally investigated the effect of different crucial processing parameters on graphite anode electrodes. Higher drying rates lead to deterioration of coating adhesion and electrochemical degradation. Further, Jaiser et al. [19] proposed a tailored tripartite drying process for graphite anodes and found that the use of mild drying conditions during the intermediate stages of drying prevents the loss of adhesion between the active layer and current collector. Recently, Altwater et al. [20] demonstrated that a three-stage drying scheme for graphite anodes can significantly reduce the drying time. However, experiments cannot provide information on the evolution of the electrode microstructure in situ or on the internal structure in a certain drying state. In addition, it is impossible to experimentally determine the drying process for all material systems.

Simulations are a strong tool for deeper process understanding and faster and easier process development. Two-dimensional (2D) models

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are commonly used. For example, Font et al. [21] proposed a mathematical model to track the evolution of binder distribution in electrodes. The model predicts that constant high drying rates are unfavorable and result in the accumulation of binder near the surface and depletion near the current collector. Lyu et al. [22] used the discrete element method (DEM) to develop a 2D model to study the evolution of the electrode structure during the drying process. Kumberg et al. [23,24] proposed a mathematical model to study the drying of electrodes of different thicknesses. The model ability to capture the change in the solvent loading of the film during the drying process was experimentally verified. Compared with 2D models, three-dimensional (3D) models can better characterize the changes in the electrode structure. For example, Nikpour et al. [25] used a multiphase smoothed-particle model to simulate the drying and calendaring of electrodes during production and concluded that drying causes inhomogeneous distribution of electrode components. Lombardo et al. [26] proposed a physics-based 3D molecular dynamics model to simulate the migration of additives during the drying process. They found that the migration of additives does not occur during the entire drying step, but rather in a specific time interval. Yao et al. [27] used coarse-grained molecular dynamics (CGMD) to simulate the microstructural evolution and debonding during drying of water-based cathode electrodes. Compared to CGMD, DEM can more explicitly account for the CBD phase in the model [26,28]. Lippke et al. [28] proposed a DEM-based model to study the drying process and captured the evolution in electrode thickness during the drying process. Hertz-Mindlin contact model was used for the contact between particles. Wolf et al. [29] proposed a drying model that combines the DEM and computational fluid dynamics and that is able to predict the thickness and porosity of the electrodes after drying. However, the above DEM-based drying models rarely explicitly consider the carbon binder domain (CBD) phase (combined carbon black and polymer binder). The spatial location of CBD affects the electrochemical properties of the electrode [30], and the complex spatial interactions between the active material (AM) and CBD phases require different contact models [31]. Therefore, 3D DEM-based drying models which considered the CBD phase will provide insights into the structural effects of interactions between the AM, CBD, and AM/CBD particles.

Calendering, as a subsequent process to drying, has a significant effect on the formation of the electrode structure [32–34]. Some researchers have experimentally investigated the effect of calendering on the electrode structure. Primo et al. [35] analyzed the calendering processability of NCM-based cathodes in terms of the AM/carbon-additive/binder composition and solvent quantity in slurry preparation. They found that higher electrochemical performances are achieved for film-like structures with higher carbon additive/binder content. Kang et al. [36] investigated the effect of calendering on the geometric characteristics and electrochemical performance of NCM111 electrodes and showed that calendering can increase the electrochemically active area and thus improve the rate capability. Li et al. [37] used an in-situ technique to analyze the repercussions of calendering status on the coupled electro-chemo-mechanical performances. The results show that the swelling strain and the stress of the Si-based electrode and the calendering degree are positively correlated. Westphal et al. [38] used a two-point approach to determine the effect of calendering on the relative resistivity of NCM cathodes and concluded that a more intimate network of particles enables shorter electron paths and more electrical contacts. Lu et al. [39] investigated the different microstructural evolutions of NCM cathodes under incremental calendering, using in-situ X-ray nanocomputed tomography, and found that electrodes comprising small particles are structurally stable with more homogeneous deformation and lower tortuosity. However, the electrode microstructure evolution is difficult to characterize in situ via experimental methods.

Some researchers applied DEM modeling to study the calendering of electrodes [40,41]. Lundkvist et al. [42,43] developed a DEM simulation model to link the local mechanical behavior of the positive electrode active layer to its global mechanical properties after calendering. They

found that the mechanical properties of the electrodes are influenced by the structure and require a deeper analysis of the structural evolution. Giménez et al. [44,45] characterized the electrode structure from the micro- to macro-scale by the developed DEM model and nano-indentation measurements. The changes in the bond number during calendering were also investigated in detail and quantitatively. Sonzogni et al. [46] studied the effect of calendering on the microstructure and electronic properties of the electrodes by DEM. The bonding structure induced by calendering involves mostly vertical tensile contacts and horizontal compressive contacts. Sun et al. [47] developed a finite element model to study the evolution of electrode microstructure and mechanical properties during calendering. The microstructure of the electrode under different calendering degrees affects the electrochemical properties. Xu et al. [48] investigated the effect of calendering on the effective cathode transport properties using a multiscale approach. The cell performance increased with increasing calendering. Zhang et al. [49,50] investigated the structural mechanics and microstructural evolution of porous electrodes during calendering using DEM, and found that increasing the number of contacts between particles can improve the conductive path within the electrode. Ge et al. [51,52] used the DEM to investigate the evolution of the electrode structure under different calendering conditions and used the bonding model to describe the mechanical response of the particles. The simulation results revealed the inhomogeneous stress distribution and anisotropic properties of electrode structures under different calendering conditions. Ngandjong et al. [53] used the DEM to study the effect of calendering on the mesostructure of the electrodes, and the effects of CBD distribution and drying were considered in the model. However, in the DEM-based simulations described above. The initial state of calendering rarely considers the structure of the drying process. Drying and calendering are unavoidable continuous processes in the production chain of electrode manufacturing [53], and randomly generated DEM models that do not consider the final structure state of drying are inappropriate for calendering simulation analysis.

In this study, a 3D representative volume element (RVE) of electrodes composed of AM particles, CBD particles, solvents, and different particle contact types was established. The structural evolution of the RVE during drying and calendering was studied sequentially and simultaneously using the developed DEM. By analyzing the RVE velocity field, coordination number, contact network, porosity, and stress, the evolution of electrode structure was uncovered. Based on this, a three-stage drying scheme was proposed, and the mechanical integrity and electrical conductivity of the electrodes were determined to be significantly improved by calendering. These results provide a promising method for the continuous study of electrode structure evolution during drying and calendering.

2. Methodology

2.1. DEM modelling method

DEM is a numerical model used to simulate the dynamics of discrete and interacting particles [54]. In DEM, a single particle's translational velocity $\mathbf{u}_i$ and rotational speed ω_i are obtained by balancing Eqs. (1) and (2):

$$m_i \frac{d\mathbf{u}_i}{dt} = \mathbf{F}_c + \mathbf{F}_i \quad (1)$$

$$I_i \frac{d\omega_i}{dt} = \mathbf{F}_t \times \mathbf{R}_i \quad (2)$$

where m_i and I_i are the i^{th} particle's mass and moment of inertia, respectively. $\mathbf{F}_c$ is the contact force between adjacent particles, $\mathbf{F}_t$ is the tangential contact force between adjacent particles, $\mathbf{F}_i$ is the force exerted on the particle, and $\mathbf{R}_i$ is the radius of the particle.

2.1.1. Contact forces of particles

When two particles in the RVE are in contact, a contact force is generated. The contact force $\mathbf{F}_c$ can be divided into [54,55]:

$$\mathbf{F}_c = \mathbf{F}^l + \mathbf{F}^d \quad (3)$$

where $\mathbf{F}^l$ provides elastic (no tension) and friction behaviors, and $\mathbf{F}^d$ provides the energy dissipation during contact. To perform numerical calculations, they were resolved into normal and shear forces as follows:

$$\mathbf{F}^l = -F_n^l \mathbf{n}_c + \mathbf{F}_s^l \quad (4)$$

$$\mathbf{F}^d = -F_n^d \mathbf{n}_c + \mathbf{F}_s^d \quad (5)$$

where $\mathbf{n}_c$ is the unit normal vector of the contact plane, and $\mathbf{F}_s^l$ and $\mathbf{F}_s^d$ in the tangential direction, respectively. The F_n^l is calculated as follows [46]:

$$F_n^l = (F_n^l)_0 - k_n (\Delta \delta_n)^{\frac{3}{2}} \quad (6)$$

where $(F_n^l)_0$ is the normal force at the beginning of the time step, k_n is the normal stiffness and $\Delta \delta_n$ is the increment of the normal relative displacement at the contact. The F_s^l is calculated as follows:

$$F_s^l = (F_s^l)_0 - k_s \Delta \delta_s \quad (7)$$

where $(F_s^l)_0$ is the linear tangential force at the beginning of the time step, k_s is tangential stiffness, and $\Delta \delta_s$ is the relative shear-displacement increment at the contact. F_n^d and F_s^d are calculated as follows:

$$F_n^d = (2\beta_n \sqrt{m_c k_n}) \dot{\delta}_n \quad (8)$$

$$F_s^d = (2\beta_s \sqrt{m_c k_s}) \dot{\delta}_s \quad (9)$$

where β_n and β_s are the coefficients of influence of the normal and tangential particles, respectively, on the contact surface. $\dot{\delta}_n$ and $\dot{\delta}_s$ are the normal and tangential velocity of the contact point, respectively. The specific equation for m_c is given below:

$$\frac{1}{E^*} = \frac{(1-\nu_1)^2}{E_1} + \frac{(1-\nu_2)^2}{E_2} \quad (10)$$

$$\frac{1}{G^*} = \frac{2(2-\nu_1)(1+\nu_1)}{E_1} + \frac{2(2-\nu_2)(1+\nu_2)}{E_2} \quad (11)$$

$$\frac{1}{R^*} = \frac{1}{R_1} + \frac{1}{R_2} \quad (12)$$

$$m_c = \frac{m_1 \times m_2}{m_1 + m_2} \quad (13)$$

$$k_n = AE^* / L \quad (14)$$

where ν is the Poisson's ratio, E is the Young's modulus, R is the particle radius, m is the particle mass, L is the spherical center distance and A is the contact area.

2.1.2. Separation of bonded particles

During drying or calendering, the particles either bond or separate from each other numerous times. Fig. 1a shows a schematic of the two particles from the bond-to-bond failure. The bond model is used to describe the mechanical responses of the binder phase [52]. The selection of an appropriate contact model from numerous available options is of importance in DEM. The linear parallel bond model (Fig. 1b) and linear contact bond model (Fig. 1c) are popular models for describing bond failure between particles [51,56].

Linear parallel bonds can limit the separation of bonded particles through the parallel-bond force $\bar{\mathbf{F}}$ and parallel-bond moment $\bar{\mathbf{M}}$. $\bar{\mathbf{F}}$ is resolved into normal force $\bar{F}_n$ and shear force $\bar{F}_s$. $\bar{\mathbf{M}}$ is resolved into twisting $\bar{M}_t$ and bending moment $\bar{M}_b$.

$$\bar{\mathbf{F}} = -\bar{F}_n \mathbf{n}_c + \bar{\mathbf{F}}_s \quad (15)$$

$$\bar{\mathbf{M}} = \bar{M}_t \mathbf{n}_c + \bar{\mathbf{M}}_b \quad (16)$$

where $\bar{F}_n > 0$ denotes tension. $\bar{F}_s$ and $\bar{M}_b$ lie on the contact plane. The force-displacement law for $\bar{\mathbf{F}}$ and $\bar{\mathbf{M}}$ consists of the following steps:

$$\bar{F}_n = (\bar{F}_n)_0 + k_n \Delta \delta_n \quad (17)$$

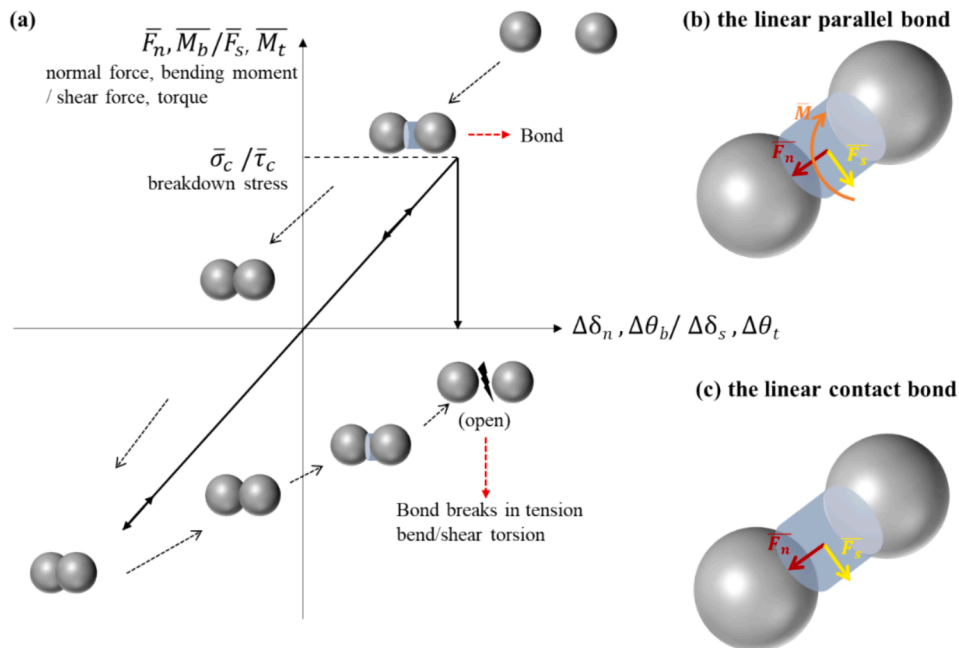


Fig. 1. (a) Schematic diagram of particles during bond-to-bond failure. (b) Linear parallel bond model diagram. (c) Linear contact bond model diagram.

$$\bar{\mathbf{F}}_s = (\bar{\mathbf{F}}_s)_0 - \bar{k}_s \bar{A} \Delta \delta_s \quad (18)$$

$$\bar{\mathbf{M}}_t = (\bar{\mathbf{M}}_t)_0 - \bar{k}_t \bar{J} \Delta \theta_t \quad (19)$$

$$\bar{\mathbf{M}}_b = (\bar{\mathbf{M}}_b)_0 - \bar{k}_b \bar{I} \Delta \theta_b \quad (20)$$

where $(\bar{\mathbf{F}}_n)_0$, $(\bar{\mathbf{F}}_s)_0$, $(\bar{\mathbf{M}}_t)_0$, $(\bar{\mathbf{M}}_b)_0$ are the values at the beginning of the time step. $\Delta \theta_t$ is the relative twist-rotation increment, $\Delta \theta_b$ is the relative bend-rotation increment. $\bar{A}$ is the cross-sectional area, $\bar{I}$ is the moment of inertia of the parallel-bond cross-section, and $\bar{J}$ is the polar moment of inertia of the parallel-bond cross-section.

The maximum normal stress $\bar{\sigma}$ and shear stress $\bar{\tau}$ of parallel bonding can be calculated from $\bar{\mathbf{F}}$ and $\bar{\mathbf{M}}$. [51,52]

$$\bar{\sigma} = \frac{\bar{F}_n}{\bar{A}} + \frac{\|\bar{\mathbf{M}}_b\| \bar{R}}{\bar{I}} \quad (21)$$

$$\bar{\tau} = \frac{\|\bar{\mathbf{F}}_s\|}{\bar{A}} + \frac{\|\bar{\mathbf{M}}_t\| \bar{R}}{\bar{J}} \quad (22)$$

The bonds are considered to be generated when the centre to centre distance of two neighbouring particles denoted by P_i and P_j meet the following criteria:

$$\|P_i - P_j\| \leq \lambda(R_i + R_j) \quad (23)$$

where λ is 1.67 [51] and R_i and R_j are particle radius, respectively. When the effect of the parallel-bond moment $\bar{\mathbf{M}}$ is not considered, the linear parallel-bond model degenerates into a linear contact-bond model (Fig. 1c). The linear parallel-bond model enables bending moments to be transferred between the two bonded particles, whereas the linear contact-bond model does not. Therefore, different contact models must be selected based on the specific physical properties of the particles.

2.2. Structural characterization during drying and calendering

The evolution of the macroscopic and microscopic structures of the electrode during drying and calendering was quantitatively characterized by the coordination number, porosity, and stress, respectively.

2.2.1. Coordination number

The connectivity of the system of active particles can be quantitatively described by the distribution of coordination numbers [49]. The coordination number C_n was calculated as follows [45]:

$$C_n = \frac{n_c^{(b)}}{N_b} \quad (24)$$

where N_b is the number of particles and $n_c^{(b)}$ is the total number of particle contacts.

2.2.2. Porosity evolution

The electrode structures were further analyzed by calculating the porosity [51]. The porosity P is defined as the ratio of the total void volume V_{void} , within the measurement region to the measurement region volume V_{reg} :

$$P = \frac{V_{void}}{V_{reg}} = \frac{V_{reg} - V_{mat}}{V_{reg}} = 1 - \frac{V_{mat}}{V_{reg}} \quad (25)$$

Where V_{mat} is the volume of material in the measurement region.

2.2.3. Stress calculation

The average stress was used to characterize the stress states in different directions during calendering [49,57]. The average stress in the volume V measurement area was calculated as follows:

$$\bar{\sigma} = -\frac{1}{V} \sum_{N_c} \mathbf{F}^{(c)} \otimes \mathbf{L}^{(c)} \quad (26)$$

where N_c is the number of contacts that lie in the measurement region or on its boundary, $\mathbf{F}^{(c)}$ is the contact force vector, $\mathbf{L}^{(c)}$ is the branch vector joining the centroids of the two bodies in contact, $\otimes$ denotes the outer product, and compressive stress is negative by convention.

2.3. RVE construction and boundary conditions

2.3.1. RVE construction of the electrode slurry

The RVE of the slurry before drying, which considers the AM particles, CBD particles, solvent, and contact between the particles, is shown in Fig. 2a. The diameters of AM particles follow the Gaussian distribution with a mean value of 13 μm [51] and standard deviation of 0.61. The CBD particles are effective particles accounting for carbon and binder and are 2.2 μm in diameter [28]. Both AM and CBD particles are assumed to be spherical for simple. The focus of this work is on the evolution of the solid components within the electrode. Therefore, direct numerical description of the fluid phase is avoided by utilizing a variety of fluid substitution models, which greatly reduces simulation time and computational cost [28]. This work uses a fluid substitution model that is common in the literature [28] and was also used in our previous work [22]. The interaction of the solvent with the particles is described by the interaction of the buoyant force F_b , the gravitational force F_g and the viscous damping force F_r [28].

$$F_b = \frac{4}{3} \pi R_i^3 \rho_{liquid} g \quad (27)$$

$$F_g = \frac{4}{3} \pi R_i^3 \rho_{solid} g \quad (28)$$

$$F_r = \alpha \times v \quad (29)$$

where ρ_{solid} and ρ_{liquid} are the densities of the particles and solvent, respectively. g is the acceleration of gravity. v is the particle motion velocity. α is the damping coefficient. Noted that the F_r is affected by the particle velocity and is not constant.

In the case study reported here [26], the slurry was composed of NCM, carbon black, polyvinylidene difluoride, and N-methyl-2-pyrrolidone. The weight ratio of AM to CBD was 94:6, with the solid component accounting for 65 %. The solid component here is the mass of AM and CBD divided by the total mass of the slurry. In the initial state, various components of the slurry were uniformly distributed in the RVE of size $200 \times 200 \times 200 \mu\text{m}$. The RVE of the electrodes before calendering, which is the state at the end of the drying process, is shown in Fig. 2b. The top pressure plate represents the roller used during calendering.

2.3.2. Boundary conditions of the RVE during drying and calendering

The initial thickness of the slurry RVE in the z-dimension is consistent with the actual slurry thickness before drying [11]. The RVE of drying exhibits periodic boundary conditions in the x- and y-dimensions. The bottom of the RVE is current collector which prevents the particles from moving down, and the top is free boundary conditions. The drying simulation ends when a stable structure is formed. The end of the drying simulation is the initial calendering condition, which meant that the electrode structure from the drying simulation is the input parameter for the calendering simulation. The RVE of calendering still has periodic boundary conditions in the x- and y-dimensions. The evolution of the structure during calendering is obtained by moving the pressure plate downward along the z-axis with a constant speed of 0.01 m/s [51].

Different contact models are selected for different types of contacts after considering the role and physical properties (CBD cohesion strength and AM-CBD adhesion strength [31]) of AM and CBD particles. The linear contact bonding model represents contacts among the AM

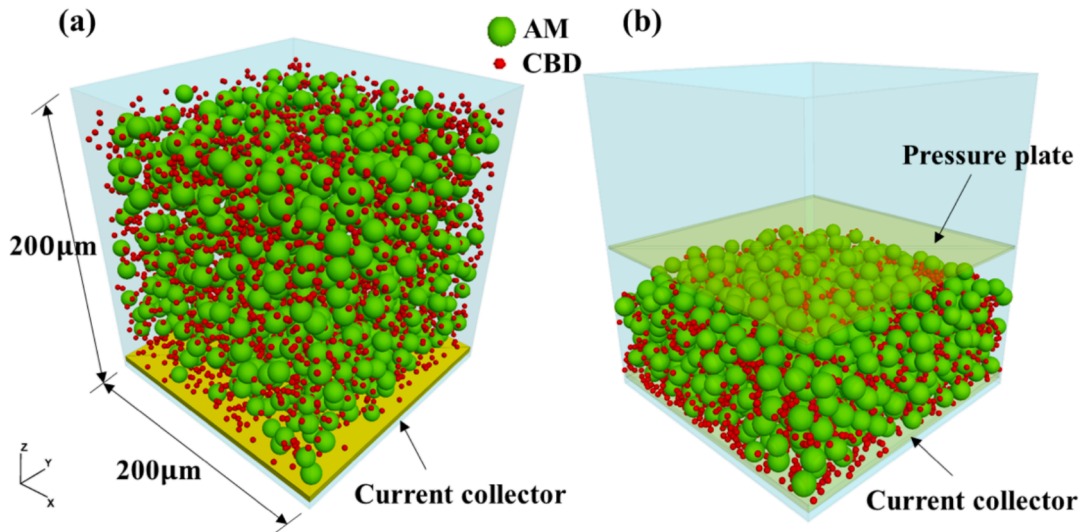


Fig. 2. Initial state of RVE during (a) drying and (b) calendaring process.

particles and between the AM particles and the current collector, and the linear parallel bonding model represents contacts among CBD particles, between CBD and AM particles, and between CBD particles and the current collector. The computational software was PFC3D version 5.00. Table 1 lists the particle characterization parameters input during the simulation. In order to validate the model, the results of the calendaring simulation were compared with the data of the electrode porosity data at different pressures measured by Giménez et al. [45] through the calendaring experiment, as shown in Fig. 3. The simulation results are in good agreement with the experimental data.

3. Results and discussion

3.1. 3D structural evolution of the RVE during drying

Fig. 4 shows the 3D structural evolution of the RVE during the drying process in different ways. For the macrostructural evolution of the RVE (Fig. 4a1-a3), the AM and CBD particles were evenly distributed in the slurry during the initial state (Fig. 4a1). The volume fraction of the particles in the slurry was small because of the high solvent content. As the drying progressed, the thickness of the slurry decreased significantly, and the volume fraction of the particles increased, which caused numerous particles to collide (Fig. 4a2). The CBD particle content at the top of the slurry was significantly increased and enriched. At the end of the drying process (Fig. 4a3), the particles were closely arranged and formed a stable porous structure. The animation of 3D structural evolution during drying can be found in video S1 (Supporting information).

The 3D structural evolution which displayed by the velocity field of the particles is shown in Fig. 4b1-b3. In the initial state (Fig. 4b1), most of the particles in the slurry remained stationary and only a few moved at a low velocity. As the drying progressed (Fig. 4b2), the bottom and middle particles of the RVE were stationary or moved at a low velocity, whereas the top particles had a larger velocity. The velocity gradient of the particles in the direction of the slurry thickness was evident. This is

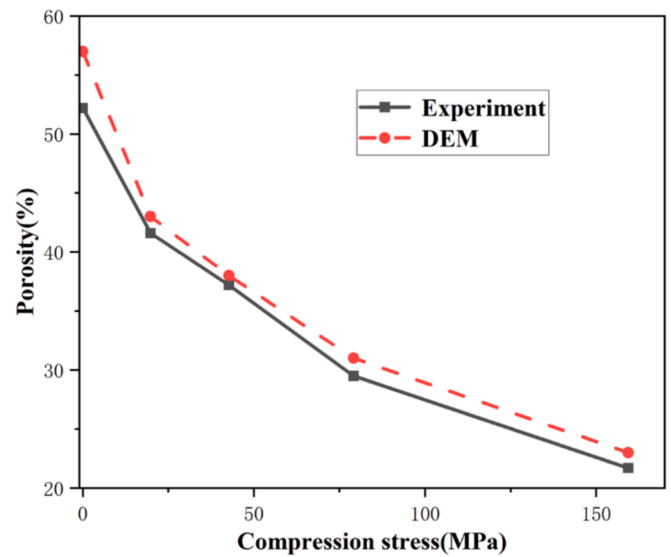


Fig. 3. Comparison of porosity between DEM simulation and experiment [45] at different pressures.

because, as the particle volume fraction increased, the effects of the particle contacts gradually increased, and the closer the particle is to the current collector, the greater the particle contact and smaller the velocity observed. Finally, all the particles remained static and a stable porous structure was formed at the end of the drying process (Fig. 4b3).

Fig. 4c1-c3 show the evolution of the contact network in the RVE during the drying process. Initially (Fig. 4c1), all the contact bonds were sparse and relatively dispersed. During the drying process at t_2 (Fig. 4c2), a contact network dominated by AM particles (thick line) began to form close to the current collector side, indicating that the porous structure began to form from the bottom of the slurry and was dominated by AM particles. The contact bonds in the RVE increased rapidly and became increasingly dense. At the end of drying (Fig. 4c3), the contact network of the RVE remained stable, forming a stable porous structure. The real time change of the contact network can be found in video S2.

The evolution of the CBD particle distribution during drying is shown in Fig. 5a-c. In the initial state (Fig. 5a), the distribution of the CBD particles was relatively dispersed in the 3D structure and uniform along

Table 1

Input parameters for particle characterization [45,49,50].

	AM	CBD
Particle density, [g cm ⁻³]	4.74	2.2
Young's modulus, [GPa]	142	0.65
Poisson ratio, [-]	0.25	0.2
Sliding friction coefficient, [-]	0.12	0.1
Rolling friction coefficient, [-]	0.012	0.01

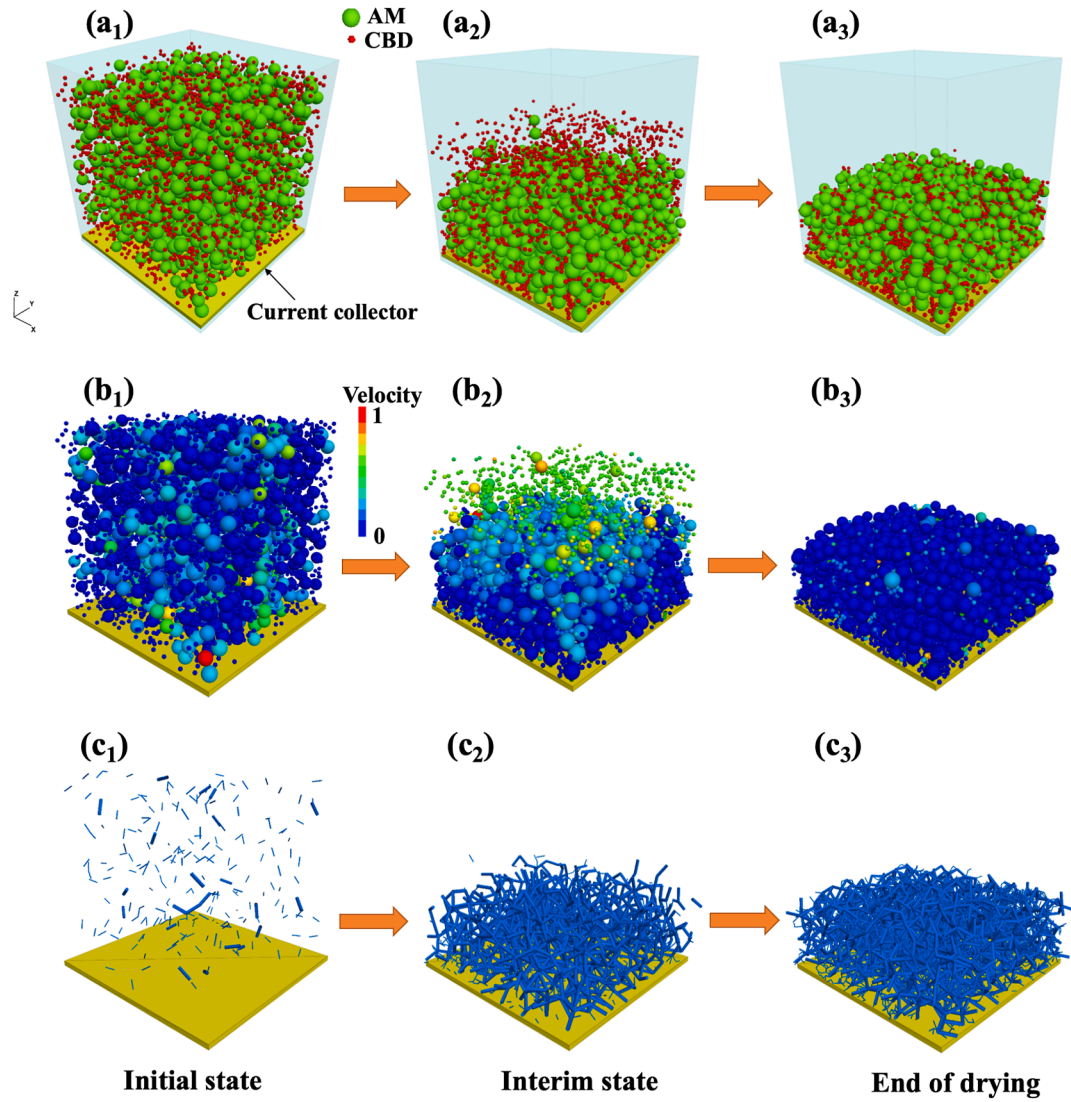


Fig. 4. 3D structural evolution of RVE during drying process. (a₁)–(a₃) Macrostructure evolution. (b₁)–(b₃) Particle velocity field. (c₁)–(c₃) Evolution of the contact network (the contact bond generated between the AM particles is represented by a thick line segment, while the other is represented by a thin line segment).

thickness direction in the 2D vertical section. During the drying process (Fig. 5b), the CBD particles in the 3D structure became denser in the interim state, reflecting an increase in the CBD volume fraction, and the distribution of the CBD particles in the thickness direction in the 2D vertical section was no longer uniform. Energy-dispersive x-ray spectroscopy (EDS) experiments also demonstrated the existence of a binder concentration gradient in the thickness direction [18,21]. At the end of drying (Fig. 5c), the distribution of the CBD particles reached the highest density in the 3D structure, and the CBD particles were enriched at the top of the 2D vertical section. This may be because CBD is more easily affected by solvents. The phenomenon of binder enrichment at the top of the electrode after drying was observed in previous EDS experiments [18], and the results of CBD enrichment at the top obtained by our simulation were consistent with the experimental phenomenon. This also reflects that the binder migration caused by drying results in a lower binder content at the interface of the coating and current collector, which reduces the adhesion.

3.2. Evolution of coordination numbers of the RVE during drying

Fig. 6 shows the evolution of the RVE coordination number during the drying process. According to the difference in the rate of increase of

the coordination number, the drying process can be divided into three stages: I, II, and III. The rate of increase in the coordination number in stages I and III was slow, whereas it was more rapid in stage II. In stage I, the particles in the slurry were relatively dispersed, and the number of particle contacts decreased at first and then increased slowly. In stage II, as the slurry thinned, particles increasingly came into contact with each other, and the number of contacts increased rapidly. In stage III, a stable porous structure was formed, and the number of contacts between the particles was also stable.

The evolution of the coordination number indicates that the rate of change in the contact number between the particles is not constant during the entire drying process, which provides a basis for optimizing the drying process. In stage II, there was significant contact between the particles, and it increased rapidly. This stage had the greatest influence on electrode formation during drying. Therefore, the drying rate was more sensitive in stage II than in stages I and III. To reduce the effect of drying rate on particle movement while accelerating the entire drying process, a three-stage drying scheme was proposed based on the evolution profile of the coordination number – a low drying rate (LDR) in stage II and a high drying rate (HDR) in stages I and III, as shown in Fig. 6. Reducing the binder content between the active layer and the current collector reduces the adhesion, and 90° peeling experiments

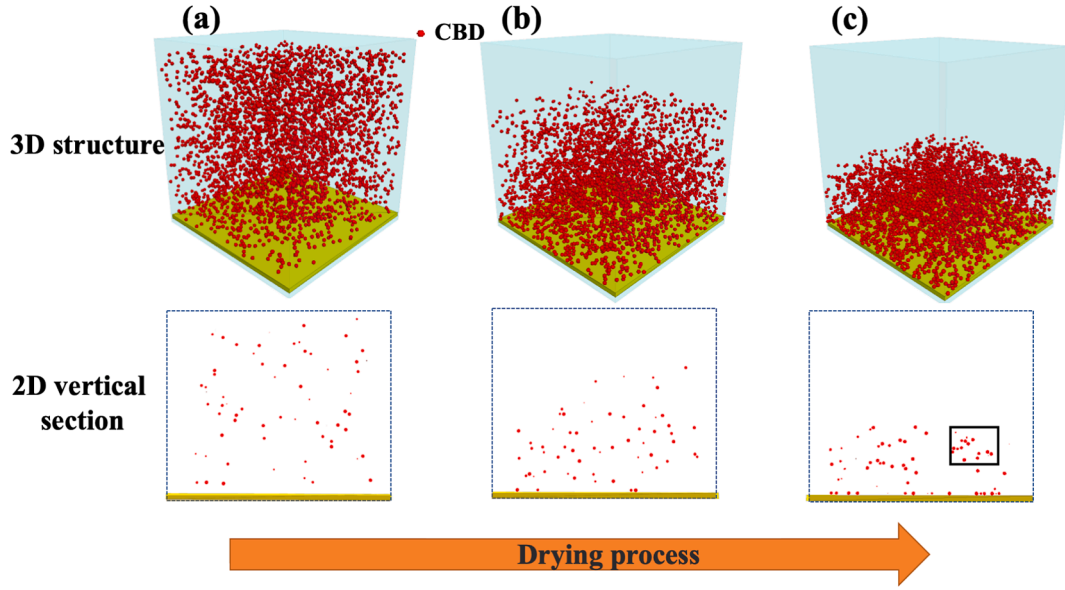


Fig. 5. (a)-(c) Evolution of CBD particle distribution during drying.

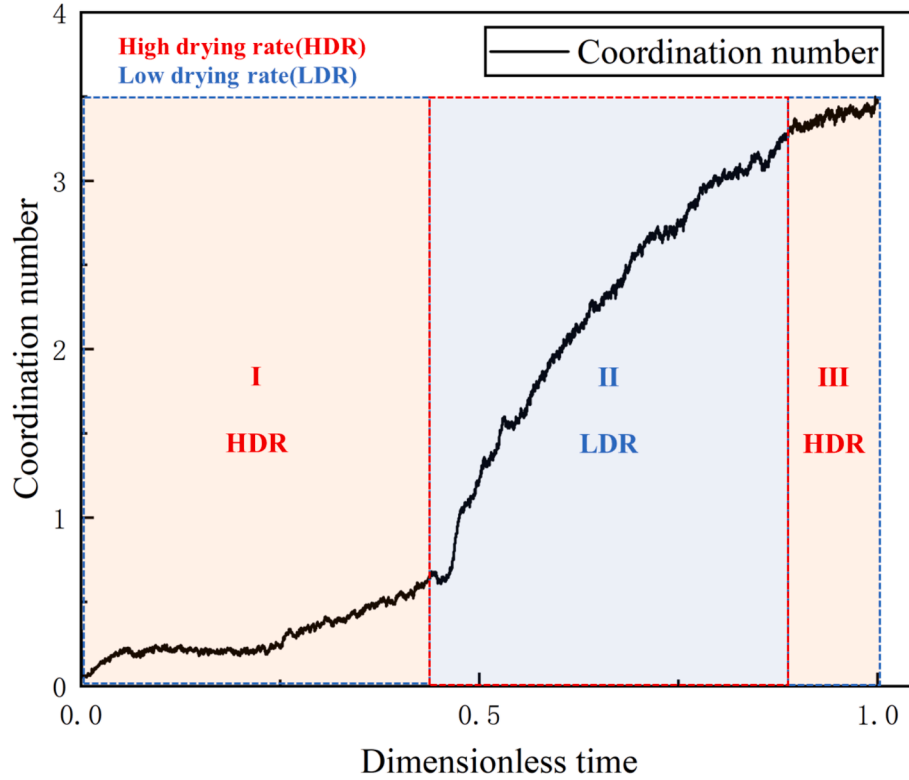


Fig. 6. Evolution of coordination numbers during the drying process and three drying stages.

have shown that using LDR in the middle of drying and HDR in the early and late stages of drying reduces the drying time while maintaining a high level of adhesion [19,20]. Qualitatively, the proposed three-stage drying scheme based on microscopic DEM simulation is consistent with the above proposed three-stage drying scheme based on macroscopic experiment. Therefore, we speculate that both have the same intrinsic mechanism. This provides a way to bridge the microstructural and macrostructural properties of the electrodes.

3.3. 3D structural evolution of the RVE during calendaring

Fig. 7 shows the structural evolution of the RVE during the calendaring process in different ways. Fig. 7a1-a3 show the evolution of the 3D and 2D macrostructures of the RVE during the calendaring process. It must be emphasized that the initial state of calendaring is the final state of the drying process (the thickness of RVE is 80 μm and the top is rough (Fig. 7a1)). During the calendaring process (Fig. 7a2), the thickness of the RVE decreased, the top became smooth, and the arrangement of the AM and CBD particles gradually became compact. In the final state

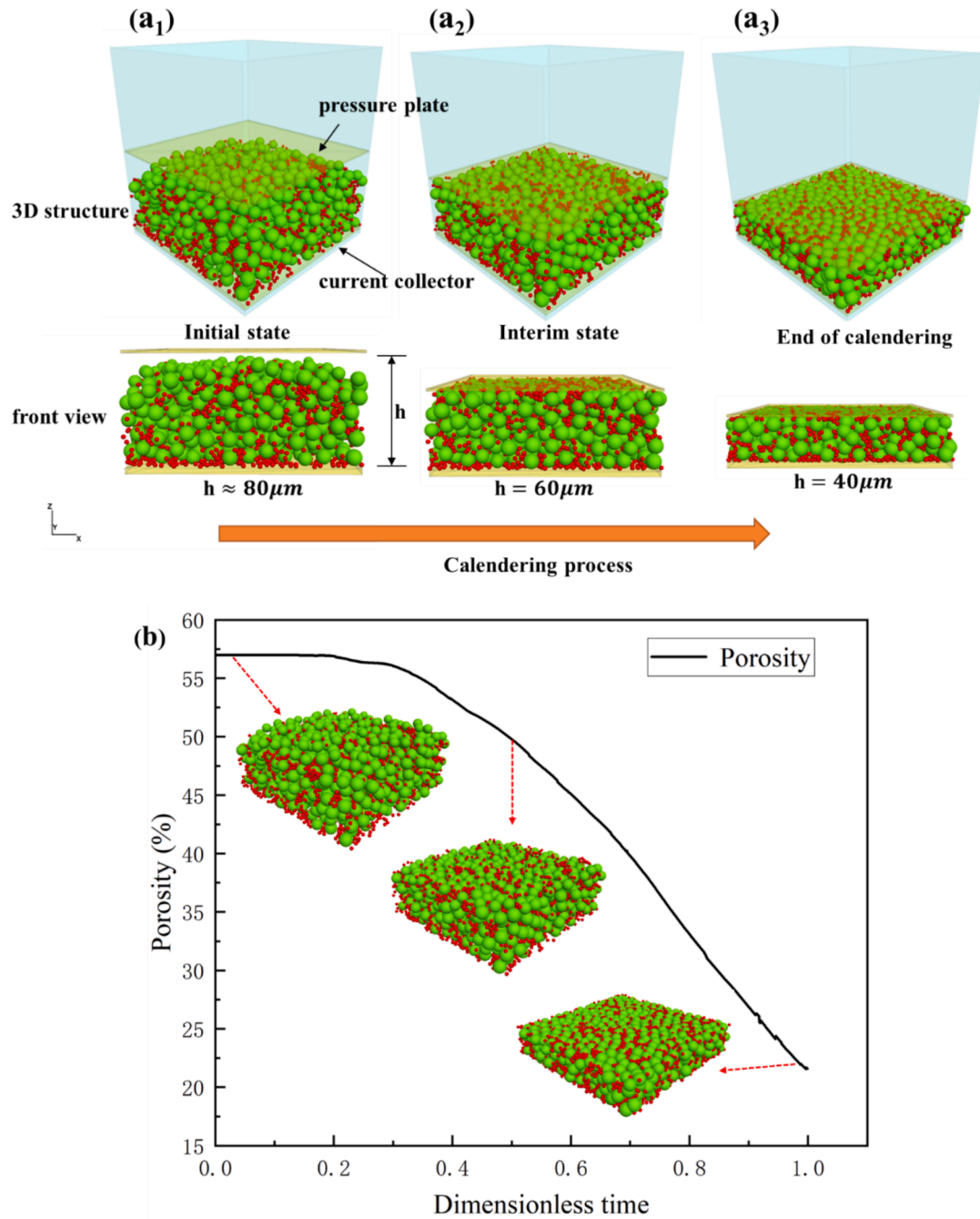


Fig. 7. Structural evolution of RVE during calendaring process. (a₁)–(a₃) 3D and 2D macrostructure evolution. (b) Porosity evolution (insets show the 3D structure of RVE at different calendaring times). (c) Coordination number evolution (insets show the 3D contact network of particles).

(Fig. 7a3), the RVE underwent remarkable transformation, as its thickness was reduced from 80 μm to 40 μm . At the same time, the density of the RVE increased significantly, indicating a more compact and densely packed structure. Experimental calendaring of the electrodes by different pressures applied to the electrodes showed that when the electrode thickness was reduced by approximately 40 % [32,33], the electrode density increased by approximately 75.4 % [32]. The comparison between the simulation results and the experimental calendaring results shows a high degree of comparability, confirming the validity of the computational models used in this study. The animated evolution of the entire 3D structure during calendaring is shown in [video S3](#).

The evolution of the electrode porosity during calendaring is shown

in Fig. 7b. The porosity of the RVE decreased from an approximate initial value of 57 % to an approximate final value of 22 % by the end of the calendaring process. The RVE in the insets shows that calendaring makes the arrangement of particles denser. The electrode volume energy density increased significantly with decreasing porosity and increasing density. The porosity of the electrodes measured experimentally by the mercury intrusion method decreased from 57 % to about 20 % [32], which is consistent with the simulated results. The change in porosity is not evident at the beginning of calendaring because the rough top of the RVE causes the pressure plate to touch only the higher points of the top layer, and this has little effect on the density of the entire RVE. During the calendaring process, the top of the RVE gradually became smooth, the influence of calendaring on the porous structure became more

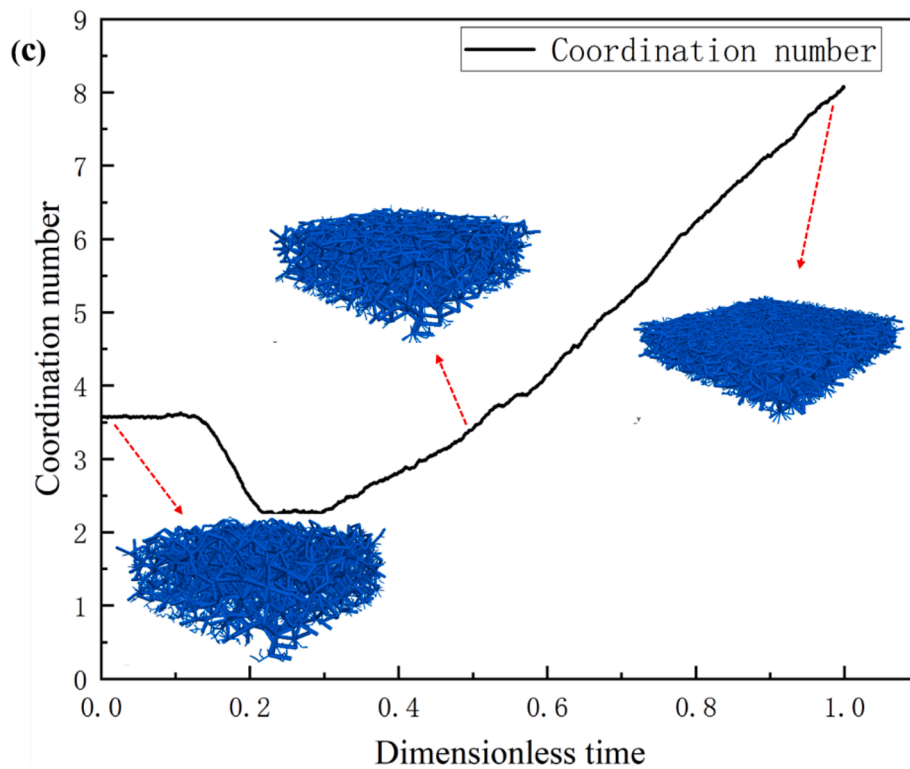


Fig. 7. (continued).

significant, and the porosity decreased linearly and rapidly until the final porosity was reached. X-ray tomography experiments on electrodes manufactured under different calendaring conditions also revealed that calendaring can produce small pores and relatively uniform pore size distribution, which in turn increases the electrochemically active area [36].

The evolution of the coordination number of the electrode particles during calendaring is shown in Fig. 7c. The coordination number of the electrode particles at the beginning of the calendaring process was 3.6 (Fig. 7c), which was the final value at the end of the drying process. The coordination number decreased slightly in the early stage of calendaring because the stability of the structure formed during the drying process

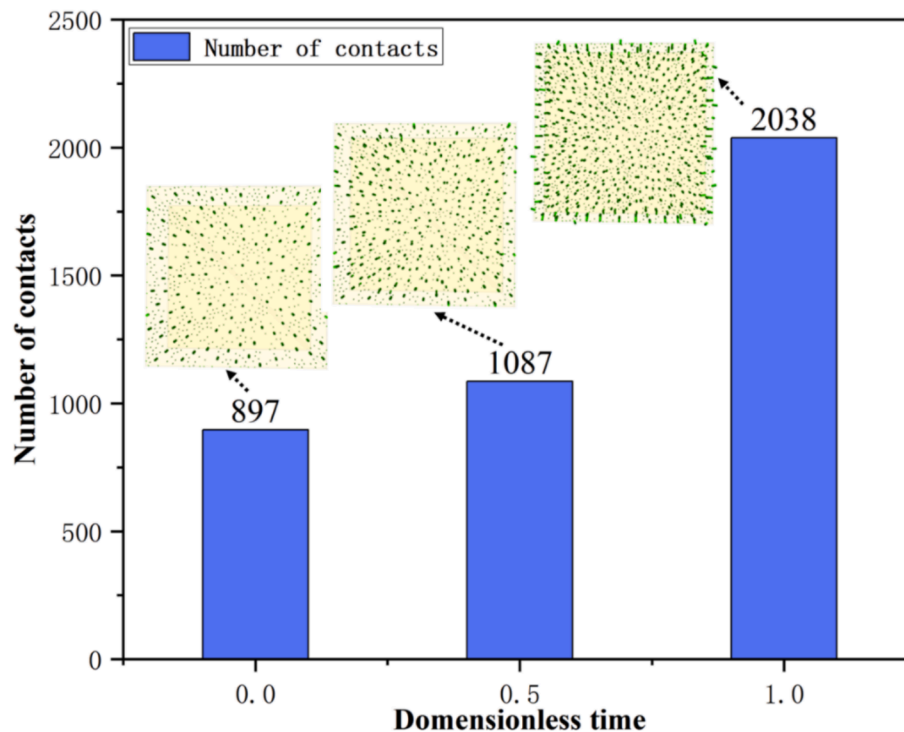


Fig. 8. Evolution of the number of contacts between the particles and the current collector during the calendaring process (insets show the contact keys on the current collector).

was destroyed and the particles that were initially in contact with each other were now separated. When the coordination number reached its minimum value, it started to increase continuously and reached a maximum value of 8.3. The large coordination number reflected increasing contact between the particles in the electrodes. Measurements of electrode resistance have shown that tighter particle networks achieve shorter electron paths and more electrical contacts effectively

reducing resistance [38]. As shown in the insets in Fig. 7c and video S4, the RVE contact network became more complete both before and after calendaring, and the particles were arranged more closely. Note that light calendaring results in reduced contact between particles, whereas excessive calendaring leads to an increase in ionic resistance [48]. Therefore, only moderate calendaring can reduce the electronic resistance and thus improve the electrochemical properties.

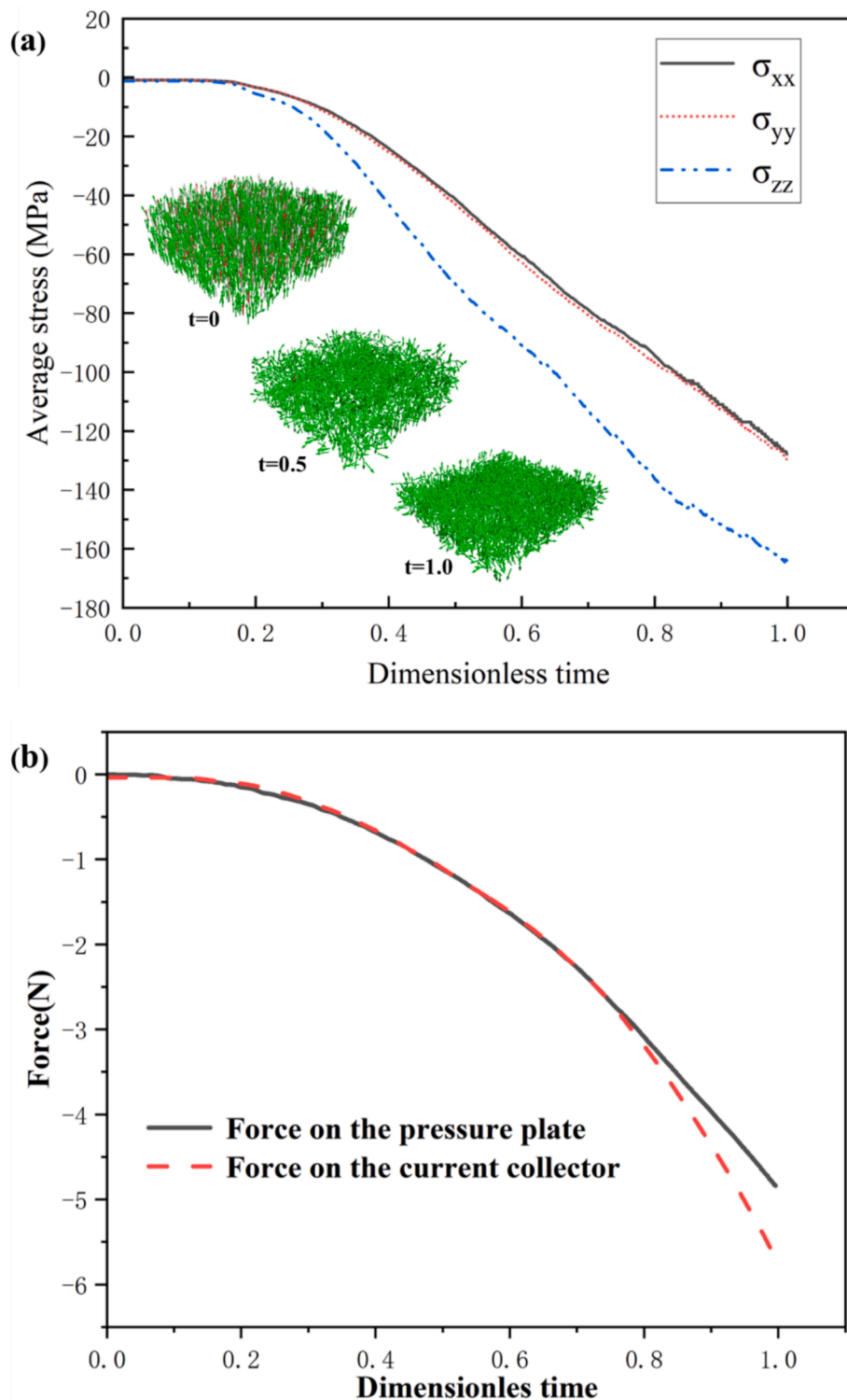


Fig. 9. Mechanical response of the RVE during calendaring. (a) The average stress of RVE changes with time, and the average stress value is negative, indicating compressive stress. (Inset shows the 3D vector diagram of the contact force on particles). (b) Change of force on the current collector and pressure plate with time in RVE during the calendaring process.

3.4. Evolution of the numbers of particle contact with the current collector during calendaring

Fig. 8 shows the changes in the number of contacts between the AM particles, CBD particles, and the current collector during calendaring. The number of contacts between the AM particles, CBD particles, and the current collector was 897 before calendaring (end of drying) and increased to 2038 after calendaring, an increase of approximately 230 %. The contact keys on the current collector showed a statistically significant increase, as shown in the inset in Fig. 8. Calendaring was able to significantly improve the contact between the AM, CBD and current collector. This improvement is manifested by a significant increase in the number of contact points, which in turn leads to two primary effects: an improvement in the interfacial adhesion between the active layer and the current collector, and a significant reduction in the interfacial resistance. Electrode resistance tests have shown that the closer contact achieved at the active layer/current collector interface is directly correlated with a reduction in contact resistance at the interface [38].

3.5. Mechanical response of the RVE during calendaring

Fig. 9 shows the mechanical response of the RVE during calendaring. Fig. 8a shows the evolution of particle stress within the RVE during calendaring. Initially, stress levels were negligible and isotropic. As calendaring progressed, the compressive stress increased linearly in all directions, with a notable increase along the z-axis due to preferential thinning of the electrodes. The equivalent average stresses along the x-y plane were maintained due to the periodic boundary conditions. The insets of Fig. 8a show the vector diagrams of the contact force on the particle, from which it is clear that the contact force is mainly along the thickness. The experiment showed that particle fracture occurred when the stress of the electrode reached 200 MPa [57]. Combined with our simulation results, it is clear that it is necessary to ensure that the stress in the direction of the electrode thickness is less than the fracture stress of the particles during calendaring.

Fig. 9b shows the evolution of the forces applied to the current collector and pressure plate during calendaring. Prior to calendaring, there were no significant forces on either component. Subsequently, under the influence of calendaring, the forces on both the current collector and the pressure plate escalated exponentially from a baseline of zero, with their magnitudes remaining approximately equivalent. This phenomenon can be attributed to the continuous decrease in porosity of the RVE, resulting in an increase in electrode density and a corresponding increase in electrode stiffness. Consequently, under conditions of constant-velocity compression, the pressure exhibited an exponential growth with time. The simulated results are consistent with previous experimental calendaring results where the pressure increased exponentially with decreasing porosity [32], demonstrating the increase in electrode stiffness during calendaring.

The force on the collector is slightly greater than the force on the pressure plate in the late stages of calendaring. This may be due to the weak local particle stability on the top of the RVE owing to the movement of the pressure plate. While calendaring can facilitate improved adhesion between AM particles and current collector [49], it is critical to recognize that excessive stress can also precipitate particle degradation [49,57]. Therefore, it is imperative to strike a delicate balance in applying pressure to ensure both interfacial adhesion and particle integrity.

4. Conclusions

In this study, a 3D RVE of electrodes containing AM particles, CBD particles, solvent, and different particle contacts was constructed by DEM. The drying and calendaring processes of the electrodes were continuously simulated. Linear contact and parallel bond models were used to characterize the different contact properties between AM particles and CBD particles, as well as the interaction between AM/CBD particles and the current collector. The structural evolution, component

distribution and mechanical response of the electrodes were analyzed in detail. The main conclusions are as follows.

1. During the drying process, there is a velocity gradient of the particles along the thickness direction. The particles near the current collector side reach stabilization first, and a contact network dominated by AM particles begins to form. Eventually, a stable porous structure is formed.
2. After drying, CBD particles were enriched at the top of the electrode. Based on the evolution of the contact state (i.e. coordination number), a three-stage drying scheme was proposed. The drying time is shortened while reducing the adverse effects on electrode formation.
3. After calendaring, the electrode thickness and porosity significantly decreased, and the coordination number and particle-current collector contact number significantly increased, which improved the mechanical integrity of the electrode. Moderate calendaring can improve the contact network of the electrode.
4. During the calendaring process, the electrode was under triaxial compression, the stress along the thickness dimension was the most significant, and the forces of the current collector and pressure plate were equal and increased exponentially with the calendaring time.

This study provides a basis for optimizing electrode manufacturing processes based on simulations that incorporate practical material properties. Since the simulation is physics-based, the modeled electrode structure can potentially be used to analyze other important properties, such as electrochemical performance and thermal conductivity. Furthermore, an in-depth evaluation of the different structures after drying in order to adjust the drying process and its influence on the subsequent calendaring process is the focus of our next work.

CRedit authorship contribution statement

Yuhang Lyu: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Shaohai Dong:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Li Ting Gao:** Writing – review & editing, Validation, Software, Formal analysis, Data curation. **Zhan-Sheng Guo:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijepes.2025.110521>.

Data availability

Data will be made available on request.

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